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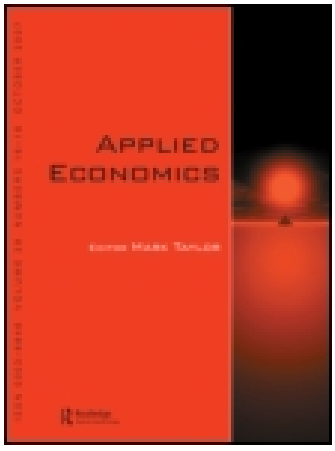

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India's petroleum demand: estimations and projections

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India's petroleum demand: estimations and projections

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Energy security is crucial for sustaining high economic growth in India. This article empirically estimates India's long- and short-term demand relations for crude oil, diesel and petrol (gasoline) using the ARDL and ECM cointegration procedures and then uses them to project demand for these products up to 2025 under various scenarios of GDP growth and oil prices. Our projections show that over 2012 to 2025, demand is likely to increase by about 74% for crude oil, 117% for diesel and 136% for petrol – the annual growth rates being about 4.3% for crude oil, 6.1% for diesel and 6.8% for petrol (gasoline). This article suggests that India needs to (1) take measures to improve efficiency in the use of petroleum products; (2) try to enhance supplies such as through production sharing agreements by Indian oil companies with other countries and (3) increase the use of nuclear, hydro, solar and other alternative energy sources, as Western European countries have done.

Keywords: demand estimations; projections; petroleum

JEL Classification: Q41; Q48

I. Introduction

Energy security is crucial for sustaining high economic growth over long periods. With rapid economic growth, petroleum demand has been rising rapidly in India. Crude oil accounts for about 29% of India's energy consumption (International Energy Agency, 2013). As India imports most of its crude oil, and there is no alternative to crude oil derivatives such as petrol and diesel for transportation and many other industrial uses, controlling crude oil consumption is difficult – and any rise in its price puts inflationary pressure on the economy. Thus, estimations of demand for crude oil, diesel and petrol and projections of likely future demand should help policy makers make appropriate supply arrangements.

Due to the global recession, crude oil prices have remained relatively stable over the past few years. The Brent crude oil spot price was US$109.64 per barrel in December 2012 and US$105.49 per barrel in December 2013 (Global Economic Monitor, 2013). But demand in the USA and Western Europe (which has been roughly constant for the past several years) could start increasing if the recession ends in a few years. Together with rapidly increasing demand from emerging economies such as China and India, this could raise crude oil prices over the long term, unless there are major new oil finds or major technological breakthroughs in alternative energy sources.

If this assessment holds, the Indian economy's growth path could slow down unless well-planned policy responses are adopted. For example, sharp

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crude oil price increases in 2007–2008 and again in 2010–2011 led to episodes of high inflation in India that slowed growth because of the tight monetary policy adopted to fight inflation. Therefore, this study empirically analyses the long- and short-term demand relations for crude oil and two of its derivatives – diesel and petrol (gasoline) – for which the demand has been rising most rapidly. We then use the long-run elasticities to project demand for these products up to 2025 for several different reasonable scenarios of GDP growth and crude oil prices.

Several previous studies estimate the demand for crude oil and other components. For India, Goldar and Mukhopadhyay (1990) and Ghosh (2009) estimate the demand for *imported* crude oil. Several studies estimate the demand for crude oil derivatives, e.g. Ramanathan (1999) and Chemin (2012) estimate the demand for petrol and Ghosh (2010) estimates the demand for diesel. Some studies estimate and forecast future demand for various derivatives of crude oil (Rao and Parikh, 1996; Ghosh, 2006; Parikh *et al.*, 2007; Kumar and Jain, 2010). Gerard and Shachmurove (2008) estimated the petroleum demand for China; Altinay (2007) for Turkey and Sa'ad (2009) for Indonesia. An interesting review of previous studies on petroleum demand estimations for various countries is available in Suganthi and Samuel (2012). Several earlier studies on India used the OLS method (Goldar and Mukhopadhyay, 1990; Rao and Parikh, 1996; Parikh *et al.*, 2007), which might yield unreliable results since most variables involved are actually nonstationary. Other studies that used cointegration techniques focused on petroleum derivatives (Ramanathan, 1999; Ghosh, 2010; Chemin, 2012) or on demand for imported oil only (Ghosh, 2009). Thus, none of these studies estimates and forecasts the *total* crude oil demand for India.[1] Even the studies that estimate imported crude oil demand (Ghosh, 2009) are dated, with data only until 2005–2006. However, since economic growth (and probably also crude oil demand) has been significantly more rapid since 2003 than earlier, it is important to estimate the demand using more recent data and provide reliable forecasts. Thus, in this study, we estimate the demand for crude oil, diesel and petrol using the ARDL cointegration technique and also project their demand up to 2025.

The rest of the article is organized as follows. Section II discusses the empirical model, econometric methodology and data sources. Section III presents the empirical results from our estimation. Section IV provides the projections of India's future demand of crude oil, diesel and petrol up to 2025. Section V summarizes the main conclusions of the article.

II. Methodology and Data Sources

Demand model

We follow the traditional demand function for crude oil,[2] where crude oil demand is simply a function of the real prices of crude oil and the real national income, which can be well proxied by real GDP at factor cost. This is appropriate since most industries using crude oil and its derivatives in various forms as the source of energy have no alternative to crude under currently available technologies for a large range of crude oil prices. While one could also estimate this relation using income per capita, given our focus on estimation and projection of India's national demand, we felt it was more convenient and expeditious to estimate the relation in terms of the national income.

This study also separately estimates the demand for diesel and petrol, two petroleum products whose demand has been rising particularly rapidly in India. We considered the possibility that other variables (such as the number of vehicles of various kinds used in India) may influence petrol and diesel demand. However, different types of vehicles consume different amounts of fuel, and consistent time-series data on the number of vehicles of various types are not available, except for a few years. Moreover, the number of various vehicles in use is itself determined by the level of economic activity. Therefore, we felt that real GDP – perhaps the single-best measure of the level of economic activity in the country – is a good explanatory variable to use instead of the number of vehicles. We also considered the possibility of including the price of substitutes, such as price of petrol, in the demand for diesel and vice versa but found it insignificant, probably because a vehicle designed to run on, say, petrol cannot be run on diesel

[1] Except Ghosh (2006); however, he used data until 2001 only and projected demand until 2011–2012. Thus, more up-to-date work is now needed given the critical importance of crude oil in the Indian economy.
[2] See, for example, Parikh *et al.* (2007), Sa'ad (2009) and Chemin (2012).

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irrespective of its price and, similarly, diesel vehicles cannot use petrol. Thus, the demand function for diesel/petrol is also estimated with real GDP and real price of diesel/petrol as explanatory variables.

We tried the linear, log-linear and double-log versions and found that the double-log model worked best for crude as well as diesel/petrol and satisfied all the diagnostic tests, such as the acceptability of functional form, normality of error term and heteroscedasticity. Thus, we used the following double-log model to estimate the demand for each of these:

$$\log D_{it} = \alpha_i + \beta_i \log P_{ti} + \gamma_i \log Y_t + u_{it} \quad (1)$$

where

D_i and P_i are the demand and real price, respectively, of the three products, where i takes the value c for crude oil, d for diesel and p for petrol;

Y is the real national income proxied here by the real GDP at factor cost;

α_i, β_i and γ_i are the parameters to be estimated and u is the error term.

Econometric methodology

As usual, we began by testing for the stationarity of the variables for the three equations (Table 1). All variables were found nonstationary in levels but stationary after first differencing, that is, they are integrated of order 1 ($I(1)$). Thus, a cointegration estimation procedure is needed. This study estimates the long-run demand for crude oil, petrol and diesel in India using the ARDL cointegration procedure proposed by Pesaran *et al.* (2001), generally recognized for reliably estimating the long-run relation.

Table 1. Unit-root tests using the ADF and PP tests

	ADF		PP	
	Levels	First difference	Levels	First difference
$\log D_{crude}$	−0.93	−3.45**	−0.51	−4.49**
$\log P_{crude}$	−1.73	−3.45**	−1.34	−5.59**
$\log D_{diesel}$	−1.37	−2.63**	−2.31	−5.84**
$\log P_{diesel}$	−0.82	−3.11**	−0.98	−6.60**
$\log D_{petrol}$	1.17	−3.33**	1.75	−3.47*
$\log P_{petrol}$	−3.57	−5.26**	−1.69	−6.66**
$\log Y$	3.21	−2.58**	4.57	−5.54**

Note: **, * indicate rejection of the null of stationarity of the series at 1% and 5% level of significance, respectively.

In estimating demand functions, one may be concerned that prices (and even the GDP) are endogenous since prices and quantities are often jointly determined (Lin, 2011). However, demand estimations obtained using cointegration procedures have the advantage that the endogeneity of the explanatory variables does not affect the reliability of results. More specifically, Pesaran *et al.* (1999) provide Monte Carlo evidence showing that the ARDL procedure that we use effectively corrects for endogeneity of explanatory variables and the estimates exhibit good small sample properties. Thus, the estimations of the demand functions obtained by us in Section III using the ARDL procedure are reliable as also the projections of the future demand in Section IV based on the estimated elasticities.

Our econometric analysis follows a two-step procedure. The first step of the procedure consists in estimating the long-run cointegration relation corresponding to Equation 1 using the ARDL procedure. The estimated coefficients β and γ represent the long-run price and income elasticities, respectively. It is expected that an increase in price, P_i, would reduce the demand for crude oil, diesel and petrol, i.e. long-run demand elasticity with respect to price would be negative ($\beta_i < 0$ for all i). On the other hand, an increase in real GDP, Y, would imply greater industrial production and increased transportation of goods and people, leading to increasing demand for crude oil, diesel and petrol, i.e. long-run demand elasticity with respect to Y would be positive ($\gamma_i > 0$ for all i).

In step two, the error-correction representation of the cointegrated relation is estimated.

$$\Delta \log D_{it} = \delta_i + \theta_i \Delta \log P_{it} + \lambda_i \Delta \log Y_{it} + \rho_i ECM_{it-1} + \varepsilon_{it} \quad (1A)$$

First differencing makes the order one integrated variables (D_i, P_i and Y) stationary, while inclusion of the ECM term (the stationary lagged residual, u_{it-1} from the long-run relation (1)) is needed to take care of the constraint imposed by the cointegrating relation among the variables at the levels (see, e.g. Pesaran *et al.*, 2001). This relation (Equation 1A) can be estimated to obtain the short-run relationship among the variables, which is the purpose of the error-correction representation of the model. The coefficient of the error-correction

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term, ρ_i, is said to be the adjustment velocity for any shock, leading to a deviation from long-run equilibrium. Furthermore, the coefficients θ_i and λ_i of Equation 1A have the interpretation of being, respectively, the short-run price elasticity and income elasticity. The short-run elasticities have the same expected sign as the long-run elasticities – negative for θ_i and positive for λ_i.

The magnitude of the short-run and long-run price and income elasticities determines the changes in demand in response to changes in the real price, and the real GDP in the short run (over a few months) and the long run (over a few years), respectively, thus, should be useful for policy makers as it has major implications for projecting the future short-run and long-run demand of crude oil, diesel and petrol in India. A long-run projection of the future demand for crude oil, diesel and petrol will be undertaken in this article. Thus, a careful empirical estimation of these elasticities will be an important focus of this study.

Data sources

This study is based on annual data for the 1970–2012[3] periods. The data for India's crude oil, petrol and diesel consumption (in million tonnes) were collected from the Ministry of Petroleum and Natural Gas, Government of India. The real GDP at factor cost (in 2004–2005 prices) is used as a proxy for real national income. The data are collected from the Handbook of Statistics on Indian Economy (Reserve Bank of India, 2013). We collected the annual average international crude oil prices of Brent oil in US$ per barrel from www.indexmundi.com and converted these into current Indian Rupees using the official exchange rate and then divided these by India's wholesale price index (WPI) (base year 2004–2005) to get the real rupee price. The Indian basket oil price is sometimes used in studies on India but that price was available only from 1980 and the two prices are essentially the same (correlation coefficient = 0.999). In the case of petrol and diesel, we used the data on their retail prices in Delhi[4] collected from the Ministry of Petroleum and Natural Gas, Government of India. We have used the diesel/petrol price in Delhi as a proxy for diesel/petrol prices in India because while prices may vary somewhat between states due to differences in local taxes and transportation costs, they are highly correlated with each other. Also, data were not available for all states for the whole sample period. We then converted the nominal price into real price by dividing it with the WPI (base year 2004–2005).

III. Results of Empirical Estimation

Demand for crude oil

We estimated the demand for crude oil in India using Equation 1 by employing the ARDL cointegration methodology. The results for the long-run demand function are given below:

$$\log D_{\text{crude}} = \underset{(4.37)^{**}}{-8.58} - \underset{(2.29)^{*}}{0.42}\log P_{\text{crude}} + \underset{(9.90)^{**}}{0.97}\log Y \tag{2}$$

F-test statistics (for testing cointegration) = 3.92**; $R^2 = .99$; $DW = 2.20$;
Serial correlation $[x^2(1)] = 2.42$; functional form $[x^2(1)] = 1.71$;
Normality test $[x^2(2)] = .72$; heteroscedasticity $[x^2(1)] = 0.003$
Note: Numbers in parenthesis below Equation 2 denote the t-statistics of the respective coefficients. Asterisks * and ** denote significance at the 5% and 1% significance levels.

The F-statistics confirm the cointegration relationship as its values are above the upper bound at the 1% significance level. Further, diagnostic tests show that serial correlation in the error term is not significant, functional form is not rejected, normality of error term is not rejected and there is no heteroscedasticity in the model. These tests corroborate the validity of the estimated demand function for crude oil. The coefficient of crude oil price is negative and significant at 5% level, indicating that crude oil price significantly influences crude oil demand in India. Further, the coefficient of income is positive as expected and statistically significant at the 1% level. The income

[3] Throughout this article, for brevity, data for a financial year, say 1970–1971 (1 April 1970 to 31 March 1971), is referred to as data for 1970.
[4] We can express the price of diesel and petrol in metric tonnes by multiplying by a constant factor (number of litres per metric tonne), but it will make no difference to the estimates of the equation – which work in percentage terms – because the double-log form is used. We used the price of diesel and petrol per litre because it is better known and understood by most people.

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elasticity is 0.97, implying that a 1% increase in real GDP would lead to an increase in the crude oil demand by about 1% in the long run. Thus, India's crude oil demand can be expected to keep increasing rapidly due to rapid economic growth.

$$\Delta\log D_{\text{crude},t} = -\underset{(1.42)}{0.72} - \underset{(2.64)^{**}}{0.04}\Delta\log P_{\text{crude},t} + \underset{(1.84)^{*}}{0.08}\Delta\log Y_t + \underset{(1.96)^{**}}{0.28}\Delta\log D_{\text{crude},t-1} - \underset{(1.94)^{*}}{0.08}ECM_{\text{crude},t-1} \quad (2A)$$

The stability of coefficients is essential in estimation for making policy recommendations. Unstable parameters can result from model mis-specification and thus lead to biased results. Since we will use the coefficients of real price and real GDP in this estimation (i.e. the price and income elasticities) for projecting future demand for crude oil, it is particularly important to test the stability of these coefficients. Stability tests like the cumulative sum of recursive residuals (CUSUM) and their square (CUSUMSQ) proposed by Brown *et al.* (1975) are considered important to analyse the stability of the coefficients. In our case, neither the CUSUM nor the CUSUM square test statistics exceed the bounds of 5% levels of significance (Fig. 1). Thus, the estimated demand relation (Equation 2) for crude oil appears stable and correctly specified.

The result of the estimation of the error-correction model, used to estimate the short-run elasticities is presented in Equation 2A. Short-run elasticity signs are in alignment with the theory – that is, an increase in crude oil price reduces demand and a higher income increases demand. With a short-run elasticity of –0.04, crude oil demand is found to be inelastic with respect to price in the short run. Similarly, the income elasticity is found to be positive and small at only 0.08, implying that the demand for crude oil is insensitive to income changes as well. As expected, short-run price and income elasticities are also found to be lower than the long-run elasticities. These results suggest that in the short run there will be very limited adjustment in crude oil demand in response to changes in price or income.

Demand for diesel

Next, we estimated the diesel demand function. The long-run estimation results are given below:

$$\log D_{\text{diesel}} = -\underset{(10.93)^{**}}{15.03} - \underset{(3.93)^{**}}{0.71}\log P_{\text{diesel}} + \underset{(11.79)^{*}}{1.20}\log Y \quad (3)$$

F-test for cointegration = 11.73**; R^2 = .99; $DW = 2.13$;
Serial Correlation [$x^2(1)$] = 1.74; functional Form [$x^2(1)$] = 3.57;
Normality test [$x^2(2)$] = 0.83; heteroscedasticity [$x^2(1)$] = 0.006
Note: Numbers in parenthesis below Equation 3 denote the *t*-statistics of the respective coefficients. Asterisks * and ** denote significance at the 5% and 1% significance levels.

The *F*-test confirms the long-run relationship between diesel price and real income on diesel consumption in India. As with the crude oil estimation, the diagnostic tests corroborate the validity of the estimated demand function for diesel. Finally, as discussed in the case of crude oil, CUSUM test is undertaken (Fig. 2) and the demand relation for diesel appeared to be stable.

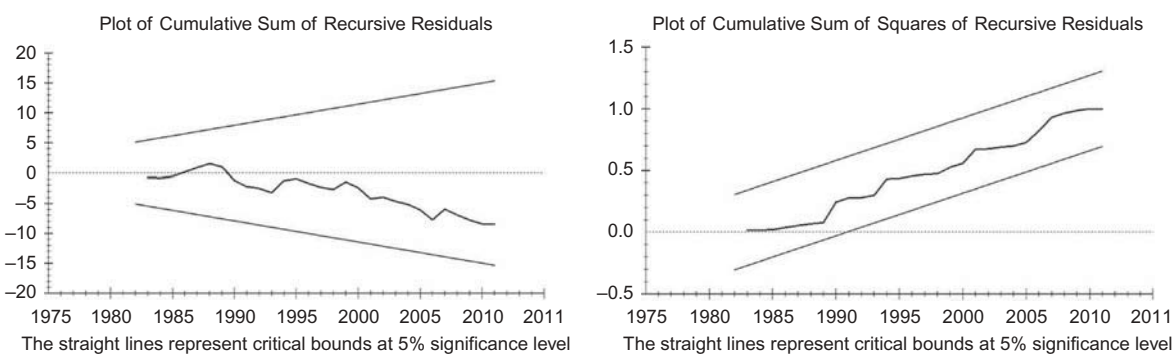

Fig. 1. Plots of CUSUM and CUSUM square statistics for crude oil demand estimation

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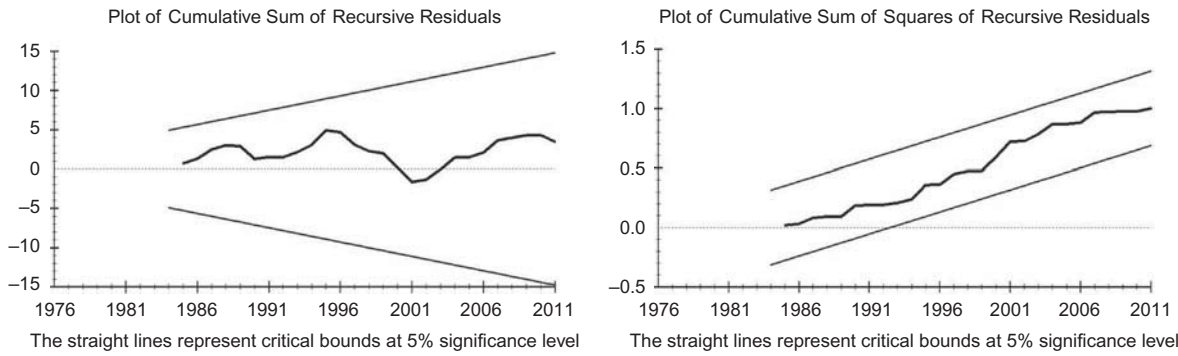

Fig. 2. Plots of CUSUM and CUSUM square statistics for diesel demand estimation

Thus, the estimated demand relation (Equation 3) for diesel appears stable and correctly specified.

$$\Delta \log D_{\text{diesel},t} = -2.30 - 0.11\Delta \log P_{\text{diesel},t} + 0.18\Delta \log Y_t - 0.16ECM_{\text{crude},t-1} \tag{3A}$$
$$(3.38)^{**}\ (3.91)^{**} \qquad (3.55)^{**} \qquad (3.63)^{**}$$

The estimation results for demand for diesel in Equation 3 show that, as expected, the price elasticity is negative and significant at the 1% level while the income elasticity is positive and significant at the 1% level. The price elasticity for diesel is −0.71, and the income elasticity is +1.20, meaning a 1% increase in real income (i.e. real GDP) leads to an increase in the diesel demand by 1.20%. Thus, the demand for diesel is quite sensitive to its price in the long run. Further, the demand for diesel is quite responsive to changes in real GDP as the coefficient is greater than 1. This could be capturing effects such as an increasing number of motor vehicles for personal travel and increased transportation of goods and services as the real GDP increases.

The result of the estimation of the error-correction model for short-run elasticities is presented in Equation 3A. Again as in the case of crude oil, results are in alignment with the theory. The short-run price elasticity is found to be −0.11 as compared to the long-term elasticity of −0.71, showing that the response to a price increase will be much smaller in the short run than in the long run (since in the long run people have more alternatives to respond to price increases than in the short run – these include switching to more fuel-efficient cars or alternative modes of transportation). Similarly, the short-run income elasticity is found to be positive and significant with a value of 0.18, significantly lower than the long-run income elasticity of 1.20.

Impact of diesel subsidy reduction

In India, diesel is subsidized relative to petrol since it is partly used by the trucking industry to transport most goods of common use. However, the prevailing 40% difference in price is neither viable nor efficient, especially as the budget and current account deficits are large. If we consider phasing out of this subsidy – so that the price of diesel rises by about Rs. 10 or about 20% per litre – given the long-run price elasticity of −0.71, the demand would decline by about 14% or about 10 million metric tons on an expected demand of about 72 million tons over the next year. This is a substantial decrease in the demand of diesel, and worth about $8 billion at the approximate current price of about $800 per ton. Thus, increasing the price of diesel would reduce India's current account deficit by about $8 billion in each subsequent year. A price increase of Rs. 10 per litre on the remaining compressed consumption of about 62 million metric tons (or about 70 billion litres) of diesel will earn the government a revenue of Rs. 700 billion (or about $12 billion) in each subsequent year, which is about 15% of the current fiscal deficit and nearly sufficient to meet the entire cost of the food security bill! On the other hand, a sharp rise in diesel prices would add to inflationary pressures immediately even though the two conflicting pressures would cancel each other

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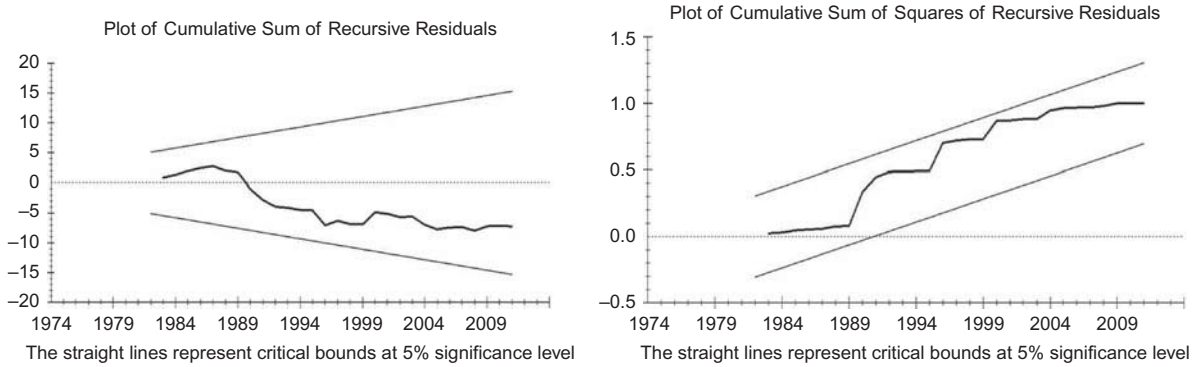

Fig. 3. Plots of CUSUM and CUSUM square statistics for petrol demand estimation

over a year or so. The temporary problem of a possible increase in inflation in the short run can be resolved by increasing the diesel price gradually by say 2–3% per month. Such a policy will cut the current account and budget deficits without adding significantly to inflationary pressures in the economy.

Demand for petrol

Finally, we estimated the demand function for petrol, and the results for the long-run relation were as follows:

$$\log D_{\text{petrol}} = -19.42 - 0.86\log P_{\text{petrol}} + 1.40\log Y - 0.66D74 \tag{4}$$
$$(23.39)^{**}\ (2.45)^{*} \qquad (15.54)^{*} \quad (1.90)^{*}$$

F-test statistics for testing for cointegration = 2.81**; R^2 = .99; DW = 1.93;
Serial correlation [$x^2(1)$] = 0.06; functional form [$x^2(1)$] = 1.88;
Normality test [$x^2(2)$] = 0.15; heteroscedasticity [$x^2(1)$] = 0.44
Note: Numbers in parenthesis below Equation 4 denote the t-statistics of the respective coefficients. Asterisks * and ** denote significance at the 5% and 1% significance levels.

The F-test confirms the existence of a long-term relation between the demand of petrol with its price and with real GDP. Further, diagnostic tests show that serial correlation in the error term is not significant, functional form is not rejected, normality of error term is not rejected and the heteroscedasticity test is at an acceptable level. These tests corroborate the validity of the estimated demand function for petrol. We included dummy variables for a few years (such as 1973, 1980, 1985, 2000, etc.) when there were large exogenous shocks to crude oil market and only D74, the dummy for 1974, was found to be significant (D74 takes the value 1 for 1974 and 0 for all other years). Finally, as discussed in the case of crude oil, we also carried out the CUSUM and CUSUM square tests for the stability of estimated coefficients (Fig. 3). Neither the CUSUM nor the CUSUM square test statistics exceed the bounds of 5% levels of significance. Thus, the estimated demand relation (Equation 4) for petrol appears stable and correctly specified.

The results of estimation in Equation 4 show that the long-run elasticity of demand for petrol in India with respect to petrol price is negative and is about 0.86. The elasticity of petrol demand with respect to real GDP has the expected positive sign and is about 1.40. Thus, the demand for petrol is reasonably sensitive to its price and seems quite responsive to changes in real GDP as the coefficient is greater than 1. This could be capturing effects such as an increasing number of motor vehicles like motor bikes, cars, etc. for personal travel as well as the increased demand for transportation of goods and services as the real GDP increases.

The short-run elasticities, estimated using the error-correction model, are presented in Equation

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4A for comparison. The short-run elasticities are smaller than their long-run counterparts. This is not unusual in the case of such goods, whose long-term responses are more definite than the short-term responses. The smaller short-run elasticities can also be explained by the slow adjustment pattern of use of automobiles by households.

$$\Delta \log D_{\text{petrol},t} = -5.24 - 0.22\Delta \log P_{\text{petrol},t} + 0.53\Delta \log Y_t + 0.16\Delta \log D_{\text{petrol},t-1} - 0.27ECM_{\text{petrol},t-1} \quad (4A)$$
$$(3.06)^{**}\ (2.75)^{**} \qquad (2.86)^{**} \qquad (1.27)^{**} \qquad (3.15)^{*}$$

These empirical estimations of demand functions for crude oil, diesel and petrol can be used to project their demand for various likely scenarios regarding real GDP growth and crude oil prices. Such projections should help policy makers know likely future demand and design supply side policies to meet it. The next section attempts such projections up to 2025.

IV. Projections of Demand for Crude Oil, Diesel and Petrol

This section projects demand for crude oil, diesel and petrol in India until 2025 using the estimated long-run demand elasticities in relations (2), (3) and (4) in Section III. The growth rate for demand for crude oil, diesel and petrol can be obtained using our demand relation (1):

$$\log D_{it} = \alpha_i + \beta_i \log P_{it} + \gamma_i \log Y_t + u_i$$

Differentiation of the demand equation with respect to time yields the relation (where a caret (^) over a variable denotes its growth rate).

$$\hat{D} = \quad \beta_i \hat{P}_i + \gamma_i \hat{Y} \quad (5)$$

which can then be used to project future demand starting from a base year, t,:

$$D_{t+1} = D_t + \beta_i P_i + \gamma_i Y_t \quad (6)$$

Using Equation 6, the future demand for crude oil, diesel and petrol can be projected using the likely scenarios for growth rates of real GDP and the real price of crude oil, petrol and diesel, respectively. Thus, for projecting demand, we need to know the likely future growth rates of real GDP and real prices of crude oil, diesel and petrol.

Likely growth rates of GDP and petroleum prices

The annualized average growth rate of real GDP between 1970 and 2012 was about 5.46%. Following the economic reforms in 1991, it averaged about 6.86% or almost 7% between 1991 and 2012, and about 7.82% or close to 8% over 2002 to 2011. It appears likely that (despite current pessimism) GDP will grow 6–8% annually on average until 2025. Thus, we consider three likely scenarios for average annual GDP growth rate until 2025: (1) a pessimistic scenario of 6%; (2) a normal scenario of 7% and (3) an optimistic scenario of 8%.

A likely scenario for the probable growth rates of crude oil prices is harder to determine given the high volatility of oil prices. However, between 1970 and 2012, crude oil prices grew at an average rate of 5.48% in real rupee terms (and 4.47% in real dollar terms). Besides, the annualized growth rate of crude oil prices has increased to 6.39% between 1991 and 2012 (4.24% in real dollar terms between 1991 and 2012). Considering these data, we consider three scenarios with oil prices in real rupees, assuming an average growth rate of about 4%, 5.5% and 7%, respectively. (Given the current price of crude oil of about $103 per barrel, these growth rates yielded 2025 crude oil prices of around $172, $207 and $250, respectively.)

Similarly, we observed that the average annual growth rate of diesel and petrol prices between 1973 and 2011 in real rupee terms was about 2.98% and 2.87%, respectively (or almost 3%). These rates are significantly lower than those of crude oil, because petrol and diesel prices also include refining, transportation and taxation components that have not

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risen nearly as rapidly as crude oil prices have. Also, the Government of India regulates petrol and diesel prices, and at times, subsidies were used to keep diesel and petrol prices from increasing in response to increasing crude oil prices. Given these historical growth rates, we consider three different scenarios of growth rate of diesel and petrol prices – 2%, 3% and 4% – for projecting diesel and petrol demand. Under these scenarios, we expect *real* diesel price (in 2012 price base) to grow from about Rs. 43.50 in 2012 to around Rs. 56.3, Rs. 64 or Rs. 72 per litre by 2025, while *real* petrol price can be expected to grow from about Rs. 68 to around Rs. 88, Rs. 100 or Rs. 113.3 per litre by 2025.

Projections for the future demand for crude oil in India

We can project future demand for crude oil in India using Equation 6, estimates of the long-run price and income elasticities of demand for crude oil obtained in Equation 2 and the above-mentioned scenarios for the likely growth rates for the GDP and prices of crude oil. The projections obtained by this procedure are shown in Table 2 and Fig. 4.

Table 2 and Fig. 4 show that the projected annual growth rate of crude oil demand would be 2.62–6.07%, per annum with a mean of about 4.34% per

Table 2. Projections of India's crude oil demand for 2012–2025 (million tonnes)

	Crude oil demand projection at 6% real GDP growth and at following the growth rate of crude oil prices			Crude oil demand projection at 7% real GDP growth and at following the growth rate of crude oil prices			Crude oil demand projection at 8% real GDP growth and at following the different growth rate of crude oil prices		
Year	4%	5.5%	7%	4%	5.5%	7%	4%	5.5%	7%
2012	*The actual demand for crude oil in 2012 is about 157.07 million tons*								
2015	173.34	169.19	165.12	179.91	175.64	171.46	186.66	182.28	177.97
2020	212.56	201.29	190.56	231.12	218.97	207.40	251.10	238.03	225.56
2025	260.65	239.48	219.91	296.90	272.99	250.89	337.78	310.83	285.88
Total percentage increase in projected crude oil demand from 2012 to 2025									
	65.95	52.47	40.01	89.02	73.80	59.73	115.05	97.89	82.01
Annualized growth rate of projected crude oil demand over 2012 to 2025									
	3.97	3.30	2.62	5.02	4.34	3.67	6.07	5.39	4.71

Source: Author's calculations.

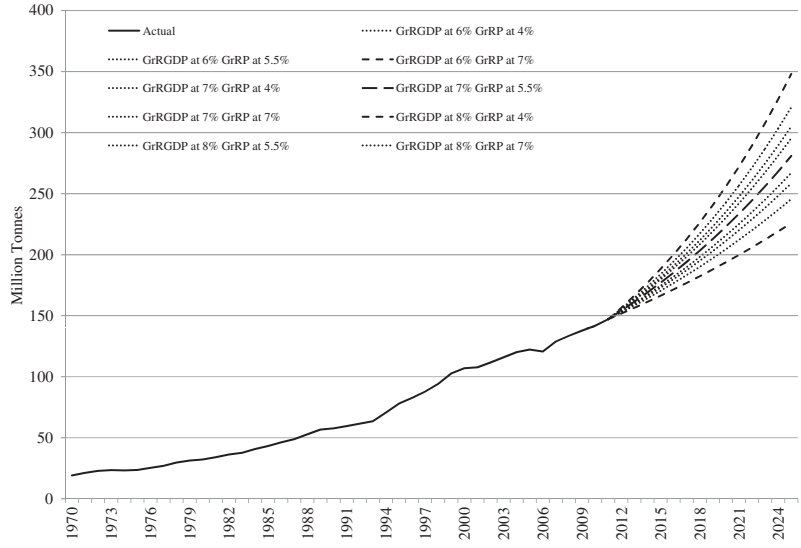

Fig. 4. Projections of India's crude oil demand for 2012–2025

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annum or about 0.6 times the assumed mean of real GDP growth rate of 7%. This will translate into a total compounded increase in crude oil demand between 40% and 115% by 2025. The average of the nine different scenarios in Table 2 is 74% for 2012–2025. For our normal and most likely case – real GDP growth at 7% and crude oil price growth at 5.5% – the projections again find a likely increase in crude oil demand from 157 million tonnes in 2012 to 273 million tonnes in 2025, i.e. an increase of about 74%. This corresponds to an average annual growth rate of about 4.34%, which implies doubling of demand in about 16 years! Thus Indian policy makers must make plans for ensuring the availability of supply for dealing with near doubling of demand in the next 16 years.

Projections for demand for diesel

Similarly, demand for diesel up to 2025 can be projected using Equation 6, long-run income and price elasticities from the estimated Equation 3 and the above-mentioned scenarios for the likely growth rates of GDP and price of diesel. The projections obtained by this procedure are shown in Table 3 and Fig. 5.

The annual growth rate of projected diesel demand is expected to be 4.08–8.20%, with a mean of 6.14% per annum, or about 0.88 times the assumed mean of

Table 3. Projections for India's diesel demand for 2012–2025 (million tonnes)

	Diesel demand projection at 6% real GDP growth rate and at following the growth rate of diesel prices			Diesel demand projection at 7% real GDP growth rate and at following the growth rate of diesel prices			Diesel demand projection at 8% real GDP growth rate and at following the growth rate of diesel prices		
Year	2%	3%	4%	2%	3%	4%	2%	3%	4%
2012	*The actual demand for diesel in 2012 is about 68.73 million tons*								
2015	79.74	77.61	75.53	83.42	81.22	79.06	87.23	84.95	82.72
2020	105.58	99.34	93.44	116.86	110.04	103.57	129.20	121.74	114.66
2025	139.79	127.16	115.60	163.71	149.07	135.66	191.38	174.45	158.93
Total percentage increase in projected diesel demand from 2012 to 2025									
	103.39	85.01	68.19	138.19	116.90	97.39	178.45	153.82	131.24
Annualized growth rate of projected diesel demand over 2012–2025									
	5.61	4.85	4.08	6.90	6.14	5.37	8.20	7.43	6.66

Source: Author's calculations.

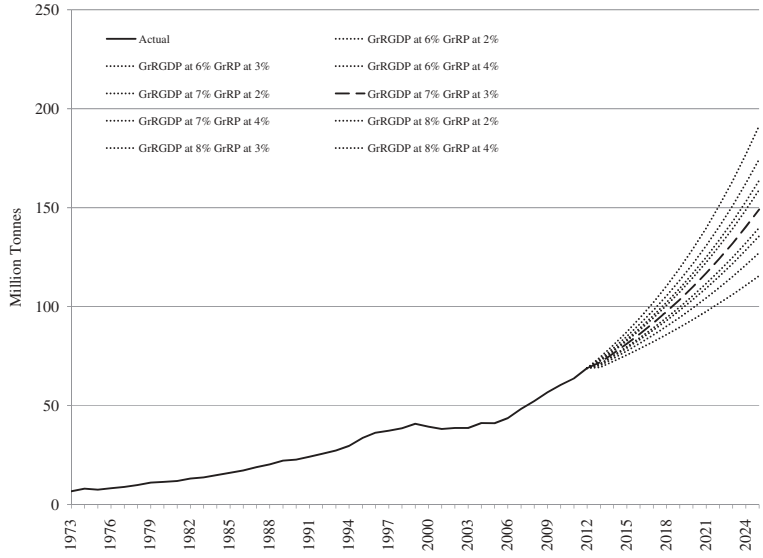

Fig. 5. Projections for India's diesel demand for 2012–2025

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real GDP growth rate of 7%. This will translate into a 68–178% total compounded increase in diesel oil demand by 2025. The average of the nine different scenarios is 117% between 2012 and 2025 (Table 3, Fig. 5). If we consider real GDP growth at 7% and diesel price growth of 3% as most likely, demand for diesel is again found to increase by about 117% (from 69 million tonnes in 2012 to 149 million tonnes in 2025 – see Fig. 5).

Projection of petrol demand in India

Finally, this section projects demand for petrol in India up to 2025 using Equation 6, using income and price elasticities from the estimated Equation 4, and the above-mentioned scenarios for the likely growth rates of GDP at 6%, 7% and 8% and price of petrol at 2%, 3% and 4% in real rupees.

The projected demand for petrol is shown in Table 4 and Fig. 6. It is seen that the annual growth rate of petrol demand would likely be in the range of 4.42–9.24% with an annual mean of 6.83%, or roughly the same as the assumed mean of the annual real GDP growth rate of 7% (Table 4, Fig. 6). This will translate into a total compounded increase in petrol demand between 76% and 215% by 2025. The average of the nine different scenarios is 136% between 2012 and 2025 (Table 4, Fig. 6). On the other hand, if we consider the case of real GDP growth at 7% and the price of petrol growing at 3% as the most likely scenario, then the demand for petrol would increase

Table 4. Projections of India's petrol demand for 2012–2025 (million tonnes)

	Petrol demand projection at 6% real GDP growth rate and at following the growth rate of petrol prices			Petrol demand projection at 7% real GDP growth rate and at following the growth rate of petrol prices			Petrol demand projection at 8% real GDP growth rate and at following the growth rate of petrol prices		
Year	2%	3%	4%	2%	3%	4%	2%	3%	4%
2012	*The actual demand for petrol in 2012 is about 16.94 million tonnes*								
2015	19.45	18.84	18.24	20.49	19.85	19.23	21.57	20.91	20.26
2020	26.90	25.03	23.28	30.25	28.17	26.23	33.96	31.66	29.50
2025	37.20	33.26	29.71	44.65	39.98	35.76	53.46	47.94	42.95
Total percentage increase in projected petrol demand from 2012 to 2025									
	119.61	96.35	75.39	163.58	136.00	111.12	215.61	182.99	153.52
Annualized growth rate of projected petrol demand over 2012 to 2025									
	6.24	5.33	4.42	7.74	6.83	5.92	9.24	8.33	7.42

Source: Author's calculations.

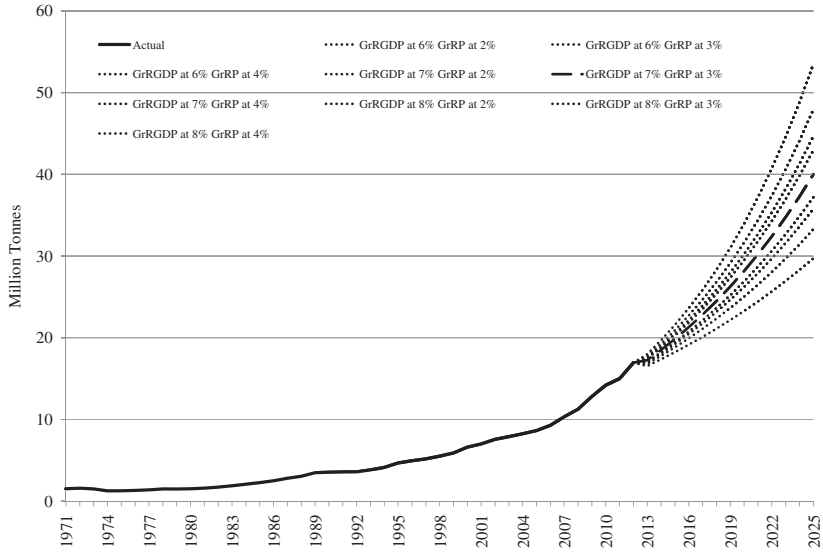

Fig. 6. Projections of India's petrol demand for 2012–2025

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from 17 million tonnes in 2012 to 39.98 million tonnes in 2025 or an increase of about 136%. The exact trajectory of projected demand for 2012 to 2025 for this case is shown in Fig. 6. Thus, although there is a large band of projections depending upon actual GDP growth and diesel price increases, petrol demand would likely increase substantially, probably around 136% or so, between 2012 and 2025.

It is clear from the above analyses that between 2012 and 2025, demand for crude oil, petrol and diesel will increase substantially by about 74%, 117% and 136%, respectively. Thus, India should take various energy efficiency and demand management measures to meet the growing demand of crude oil, petrol and diesel in the long run. This should include market-linked prices for petrol and diesel. Petroleum subsidies on fertilizers, kerosene and liquefied petroleum gas (LPG) need to be well targeted and minimized to the extent possible.

Evaluation of the predictive performance of our estimates

Model selection depends not only on the goodness of fit of a model to the data, but also on the performance of out-of-sample forecasts. In this section, we check our effort by measuring the strength of the ARDL long-run estimation model in forecasting the likely future demand for crude oil, diesel and petrol. To validate our model, we divide our data set into two parts. First, from 1970 to 2006, we re-estimate our model based on Equations 1, 2 and 3 and estimate new income and price elasticities of the crude oil, diesel and petrol for the above sub-period (Table 5). Using these sub-period elasticities and the actual annual growth rate of GDP and prices of crude oil, diesel and petrol, we forecast the respective demands for the second sub- period 2007–2012 and then compare the forecasted values with the actual values.

The actual and predicted values of the three energy variables and the percentage difference between them are shown in columns 1, 2 and 3 of Table 6. The table shows that in the case of crude oil demand, the projected value based on actual growth rates reveal upward and downward biases in different years. However, the difference is within 5% level in all years except in 2009. Similarly, in the case of diesel and petrol demand, the percentage difference between actual and predicted value is in a reasonable range of fewer than 5%.[5] Because of the international financial crisis, 2008 and 2009 were very unusual years: petrol demand declined more steeply then predicted, probably because of a sharp downward

Table 5. Long-run elasticities estimated for full and restricted period

Demand equation	Price elasticity	Income elasticity
Full period (1970–2012)		
Crude oil	−0.42	0.97
Diesel	−0.71	1.20
Petrol	−0.85	1.40
Restricted period (1970–2006)		
Crude oil	−0.34	1.00
Diesel	−0.66	1.07
Petrol	−0.82	1.38

Table 6. Predictive performance of our ARDL long-run estimation models[a]

	Actual demand	Predicted demand based on actual growth rate of GDP and prices	% difference between (1) and (2)
	(1)	(2)	(3)
Demand for crude oil (Million tonnes)			
2007	129.0	127.7	1.0
2008	133.6	137.4	−2.9
2009	137.8	150.5	−9.2
2010	141.8	146.3	−3.2
2011	147.2	140.0	4.9
2012	157.1	153.9	1.9
Demand for diesel (Million tonnes)			
2007	48.3	47.9	0.7
2008	52.3	54.1	−3.4
2009	56.7	55.8	1.5
2010	60.5	61.5	−1.7
2011	63.7	65.4	−2.6
2012	68.7	67.9	1.1
Demand for petrol (Million tonnes)			
2007	10.3	9.8	4.7
2008	11.3	11.6	−3.6
2009	12.8	12.8	−0.5
2010	14.2	14.8	−4.7
2011	15.0	16.1	−7.8
2012	16.9	15.9	5.8

Notes: [a]We estimated the long-run price and income elasticities using the ARDL model for the period 1970–2006 and use them in Equation (6) to forecast the demand for the period 2007–2012 using actual growth rates of GDP and petroleum prices.

[5] Except for petrol in 2011 and 2012, where the error still remains under 8%.

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re-appraisal of the world economy's future. The forecast performance of the models for diesel and petrol is somewhat better than that for crude oil, because in India the government regulates prices for diesel and petrol, and their movement was much more gradual than that of the crude oil prices, which are determined freely in the international markets and are much more volatile. The above results show that the predictive performance of our models is reasonably good.

V. Conclusions

This article empirically estimated long- and short-run demand relations for crude oil, diesel and petrol for India using the ARDL cointegration procedure and data from 1970 to 2012 and found long-run income elasticity of about 1 for crude oil, 1.2 for diesel and 1.4 for petrol. Thus, demand for petroleum seems to be growing about proportionately to real GDP. Further, the long-run price elasticities were found to be −0.42, −0.71 and −0.85 for crude oil, diesel and petrol, respectively, which show that demand declines moderately with rising petroleum prices. The short-run price and income elasticities were found to have the same sign as the long-run elasticities, but had much smaller magnitudes between 0.04 to 0.3 as the adjustments in the demand are much more difficult in the short run than in the long run.

Using the estimated long-run elasticities, we projected the demand for crude oil, diesel and petrol until 2025 for three possible scenarios of GDP growth rates and oil price growth rates, based partly on past trends. For the normal scenario of 7% real GDP growth rate and 5.5% growth rate of real crude oil price, crude oil demand in India would increase at the rate of about 4.34% annually, which means a total increase of about 74% between 2012 and 2025. Similarly, the demand for diesel and petrol would respectively increase by about 6.14% and 6.83% annually or by a total of 117% and 136%, respectively, between 2012 and 2025 for the normal scenario of 7% real GDP growth rate and 3% growth rate in real diesel and petrol price.

Thus, India has to brace not only for increasing crude oil demand, but also for possible increases in crude oil prices. For example, if India's crude oil demand increases by about 75% by 2025 and crude oil price too increase by 50% to 100% by 2025, India's crude oil import bill (which stood at about $155 billion in 2011–2012) would increase by a factor of 2.6 to 3.5 (to about $400 to $540 billion) by the year 2025. Considering that India's total exports were only $305 billion in 2011–2012, this is a worrisome scenario and requires urgent action.

Our analysis suggests that energy security could become a major issue for India that could hurt its ability to sustain high rates of economic growth in the future. For improving its prospects for energy security, India needs to take all possible measures for demand conservation to minimize the growing demand of crude oil, petrol and diesel, to enhance the efficiency of energy use, and to augment future energy availability. This should include market-linked prices for petrol and diesel. Petroleum subsidies on diesel need to be eliminated gradually over the next 1–2 years, and those on fertilizers, kerosene and LPG need to be well targeted and minimized to the extent possible. Vehicle producers should be required or incentivized to continuously try to increase the fuel efficiency of vehicles produced. India needs to enhance petroleum supplies through increased domestic explorations as well as other measures, such as participation in exploration and production in foreign oil fields by Indian oil companies (which the Chinese are using extensively). It should increase production and use of coal wherever possible, given its abundant coal reserves. India needs to use renewable energy sources (hydro, solar, wind, bio-fuels, nuclear, etc.) on a much larger scale as Western European countries are doing. Finally, India should also take measures to increase exports to meet its growing future oil import requirements.

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